



**KTH Microelectronics
and Information Technology**

Exam in ID2209 Distributed Artificial Intelligence and Intelligent Agents, 2023-01-11, 08:00-12:00

Rules

This exam is “closed book” and you are not allowed to bring any material or equipment (such as laptops, PDAs, or mobile phones) with you. The only exceptions are an English to “your favorite language” dictionary and pencils.

Instructions

- Please read the entire exam first!
- Write clearly
- Each sheet of paper must contain your name, ”personnummer”, Problem number and a unique sheet number
- Write only on one page of a sheet. Do not use the back side
- Only one Problem must be reported on each sheet
- If more than one sheet is needed the continuation should be clearly noted on the beginning of each sheet and the sheet numbers used should be consecutive
- Always motivate your answers. Lack of clearly stated motivation can lead to a reduction in the number of points given
- The tasks are not necessarily sorted in order of difficulty. If you get stuck it might be a good idea to go on to the next task.
- The Parts are sorted in order of difficulty but inside of the Parts problems are not sorted in the order of difficulty

Grading

1. In order to pass exam (grade E) you have to get at least 36 points from questions from Part I (this includes your bonus points).
2. In order to get grade D you have to pass exam (see item 1) and get 49 points from all questions (this includes your bonus points).
3. In order to get grade C you have to pass exam (see item 1) and get 63 points from all questions (this includes your bonus points).
4. In order to get grade B you have to pass exam (see item 1) and get 74 points from all questions (this includes your bonus points).
5. In order to get grade A you have to pass exam (see item 1) and get 85 points from all questions (this includes your bonus points).

GOOD LUCK!

Part I (48 p.)

What is an agent?

1. If traffic lights (together with their control systems) are to be considered as intelligent agents, which of agent's properties should they employ and which of agent's properties they can't employ (if any)? Illustrate your answer by examples.

(6p)

Agent theory

2. What are relations between Beliefs, Desires, Goals and Intentions in the BDI-architecture?

(4p)

Agent Architectures

3. What is a purpose of introducing the "see" function into an abstract agent architecture? Give an example

(4p)

4. What are basic types of layered agent architectures for hybrid agents?

Briefly explain their advantages and disadvantages

(4p)

Negotiation

5. Pareto efficiency is an important property in negotiation. Briefly explain it. What are relations between Pareto efficient solutions and social welfare maximizing solutions?

(4p)

6. If we compare the binary protocol and Borda protocol. Which of them is more computationally efficient? Which of them is not independent of irrelevant alternatives? Explain.

(4p)

7. We considered Clark tax algorithm. What is its objective? How it achieves this objective? (It is not necessary to present particular formulae in your answer)

(4p)

Communication

8. Having semantics of the ***request(s,h,α)*** speech act as it was defined in lecture notes, which of the following statements is/are precondition(s) and which of them is/are post-condition(s) of this speech act?

- ***h believes (s believes (s wants α))***
- ***s believes (h believes (h can do α))***
- ***s believes (s wants α)***
- ***s believes (h can do α)***

(3p)

9. Give basic ideas of the agent communication languages KQML and FIPA. What do they have in common and what are basic differences between them?

(4p)

Coordination

10. Explain main organizational structures that you know. Give their advantages and disadvantages. (4p)
11. Give examples of social norms and laws in agent coordination. What is difference between them? (3p)

Mobile agents

12. Give examples of situations (at least 2 situations) where mobile agents are appropriate (when they have benefits over communicating non-mobile agents)? Justify your answer. (4p)

Part II (25 p.)

13. What are axioms of the logic of knowledge? Which axiom(s) from the logic of knowledge is/are not valid in the logic of belief and why? (5p)
14. This is a payoff matrix for the Prisoners dilemma:

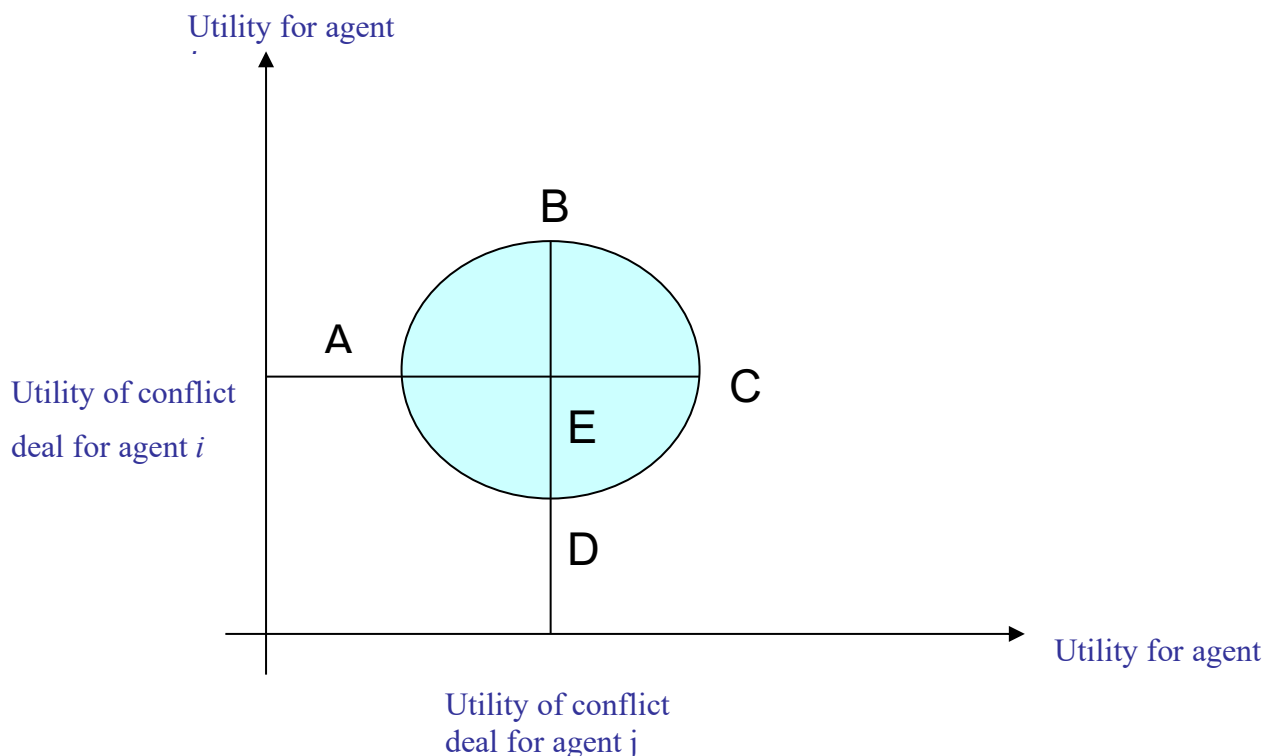
		<i>i</i>	
		Defect	Coop
<i>j</i>	Defect	2, 2	0, 5
	Coop	0, 5	3, 3

Which cells in this matrix correspond to: social welfare, Pareto optimal solution and Nash equilibrium? Explain your answer.

- (5p)
15. What is/are the main difference(s) between meta-level information exchange coordination and multi-agent planning? (5p)
16. What is/are main difference(s) between MAS architectures that use middle agents and market-based MAS architectures? (5p)
17. Show an implementation (in any (pseudo-)code) of action selection function for agents in the subsumption architecture? (5p)

Part III (24 p.)

19. We considered the following figure in the class.



Show on the figure points/areas for:

Conflict deal, negotiation set, possible deals, Pareto optimal deals and individually rational deals.
Explain your answer.

(7p)

20. Is it possible that the same propositional content of the ACL message can be used to present different speech acts? Give an example if the answer is yes and justify if the answer is no.

(7p)

21. If we assume that the statement “agent doesn’t know whether p ” is equal to “ $\Diamond p \ \& \ \Diamond \neg p$ ” in modal logic. Express in the logic of knowledge the following statement:

“I don’t know whether I know the answer and I don’t know whether my neighbor knows the answer”

Explain your answer.

(10p)

-----End of Exam-----